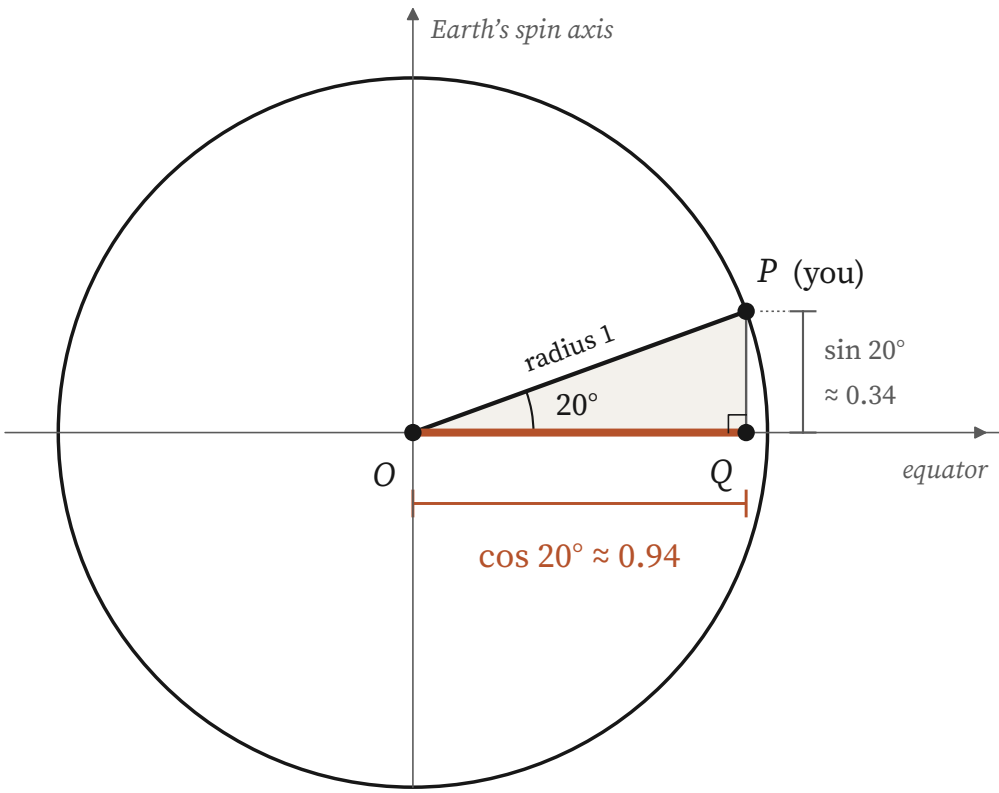
The Cosine of Latitude and Eastward Ground Speed

At latitude φ you stand a distance R cos φ from Earth's axis, so your daily circle, and your eastward ground speed, shrink as the cosine of your latitude.

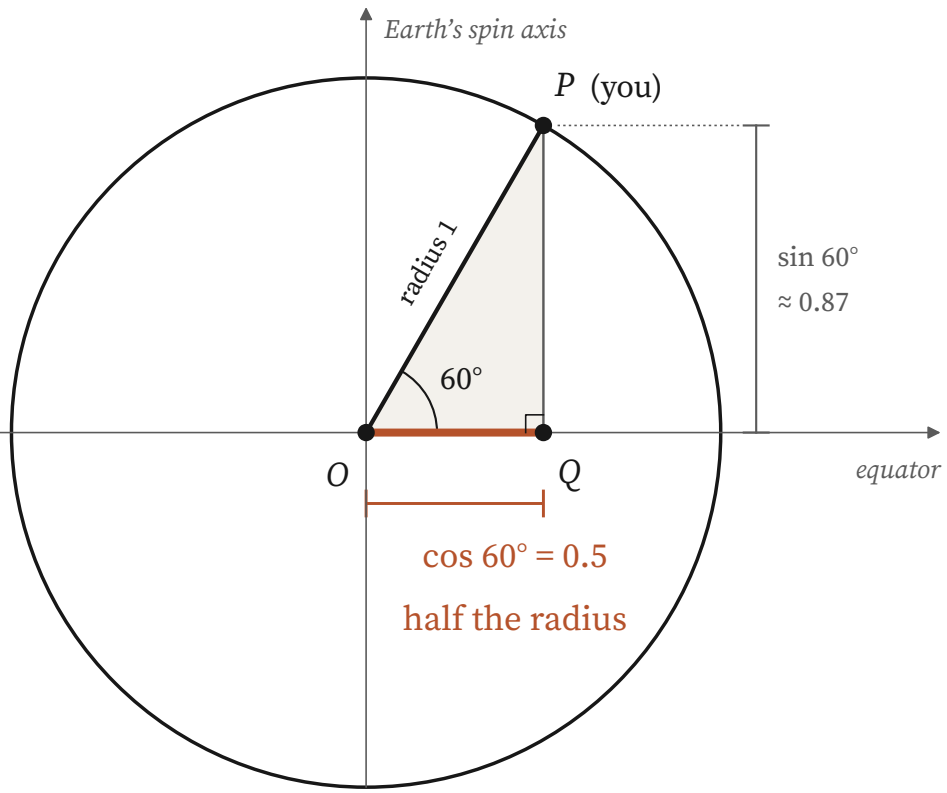
Latitude 20°



Eastward speed ≈ 0.94 × 1,040 ≈ 977 mph

Q sits near the edge, 0.94 of the way out: you keep 94% of the equator's speed.

Latitude 60°



Eastward speed = 0.5 × 1,040 = 520 mph

Q sits halfway out: you keep exactly half of the equator's speed.

Latitude	0° equator	20°	40°	60°	80°	90° North Pole
cos φ	1	0.94	0.77	0.50	0.17	0
Eastward speed	**1,040 mph**	**977 mph**	**797 mph**	**520 mph**	**180 mph**	**0 mph**

Eastward ground speed = (speed at the equator) × (cosine of latitude) $v = v(\text{equator}) \times \cos\phi = 1{,}040 \times \cos\phi$ mph

Every number in the table is 1,040 × cos φ. The 1,040 mph is the equator's 24,900 miles carried around once in a sidereal day, about 23 hours 56 minutes.